EEES-302 Digital Design

Experiment 4

Designing 2-bit Multiplier Circuits

1. Objective:

The main purpose of this experiment is to understand the working principles of combinational logic circuits by designing a 2-bit multiplier.

Additionally, it aims to provide practical, hands-on experience in physically building and wiring this mathematical circuit on a breadboard using basic logic gate ICs.

2. Review of Experiment:

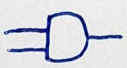
A 2-bit multiplier is a digital circuit that multiplies two 2-bit binary numbers. Assume that the inputs as $A(A_1, A_0)$ and $B(B_1, B_0)$. Because the maximum decimal value for a 2-bit number is 3 (11), the maximum possible result is $3 \times 3 = 9$ (1001). Therefore, the output requires 4 bits, which which we call P $(P_3 P_2 P_1 P_0)$.

$$\begin{array}{cccc} & & A_1 & A_0 \\ \times & & B_1 & B_0 \\ \hline & & A_1B_0 & A_0B_0 \\ & A_1B_1 & A_0B_1 & \downarrow \\ \hline P_3 & P_2 & P_1 & P_0 \end{array}$$

The experiment requires XOR and AND gates after the K-map implementation.

AND Gate

A	B	F
0	0	0
0	1	0
1	0	0
1	1	1



$F = A.B$

XOR Gate

A	B	F
0	0	0
0	1	1
1	0	1
1	1	0



$F = A \oplus B$

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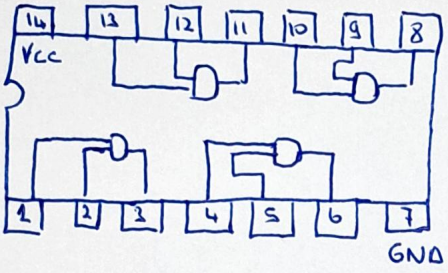
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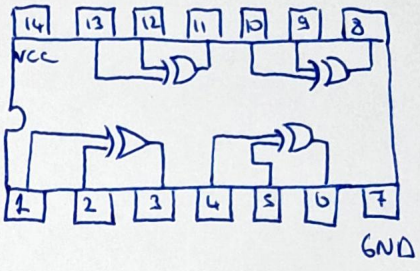
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3. Equipment List:

The experiment requires AND and XOR gate IC. 74LS08 for AND gate is used and 74LS86 is used for XOR gate. Breadboard and jumper wires also is used. DC power unit for the input terminals and Ossilator for the output terminals are used.



74LS08 AND Gate IC



74LS86 XOR Gate IC

4. Preliminary Work:

Design a combinational logic circuit with multiple outputs that will serve as a 2-bit multiplier.

A_1	A_0	B_1	B_0	P_3	P_2	P_1	P_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	0
0	1	0	1	0	0	0	1
0	1	1	0	0	0	1	0
0	1	1	1	0	0	1	1
1	0	0	0	0	0	0	0
1	0	0	1	0	0	1	0
1	0	1	0	0	1	0	0
1	0	1	1	0	1	1	0
1	1	0	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	0	0	1	1	0
1	1	1	1	1	0	0	1

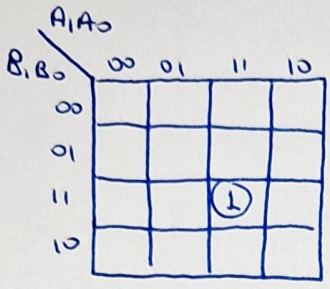
Truth table of inputs and outputs are shown in the table.

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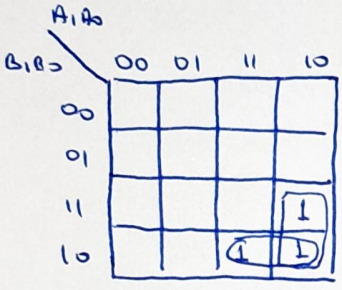
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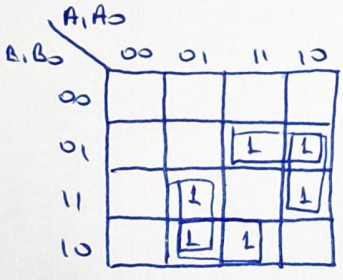
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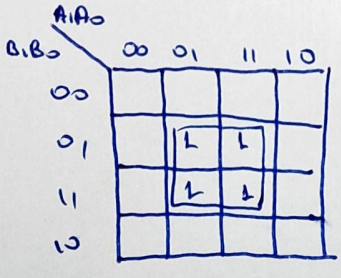
$F_3 = A_1A_0B_1B_0$



$F_2 = A_1\bar{A}_0B_1 + A_1B_1\bar{B}_0$



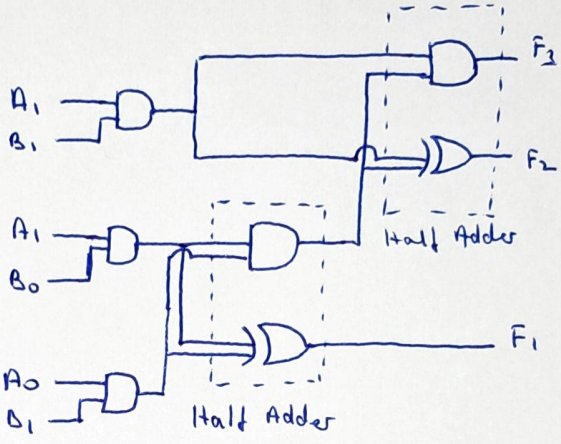
$F_1 = A_1\bar{A}_0B_0 + A_1\bar{B}_1B_0 + \bar{A}_1A_0B_1 + A_0B_1\bar{B}_0$

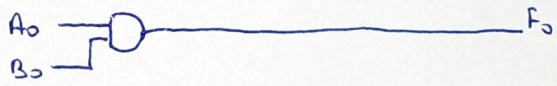


$F_0 = A_0B_0$

Circuit Implementation:

The logic gate diagram can be reduced by using two half-adders.





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5. Experimental Procedure:

Preliminary work and the experiment are the same. So, we implemented the reduced form of the circuit by using logic gates.

6. Experimental Results:

During the laboratory session, the 2-bit multiplier circuit was succesfully built on a breadboard using physical logic gate.

To verify the circuit, we tested all 16 possible input combinations and compared the outputs to the theoritical truth table.

- When the inputs were set to A=10 and B=11, the output as 0110.

- When the maximum inputs were applied, A=11 and B=11 the outputs succesfully displayed 1001.

The physical circuit outputs perfectly matched the expected theoritical results for every combinations. This confirms that the logic gates were wired correctly and the combinational logic design was succesful.

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